

Customer Analytics Final Report

Boutique Winery Sales

Team 20

Executive Summary

This report analyzes a data set from a boutique winery to improve customer engagement and targeting through several models. An RFM (Recency, Frequency, Monetary Value) model was deployed to redefine customer segmentation beyond the existing static labels. Logistic regression was utilized to predict churn and identify target customers for Winemaker Calls. Finally, linear regression confirmed an omnichannel premium in customer lifetime value, demonstrating that engaging customers across multiple channels is as valuable as retaining them for three additional years. Our analysis resulted in several recommendations for the winery, including using the Tasting Room as a recruitment anchor and revamping email content to reduce attrition.

Data Overview

Thanks to the good people at Freaklytics, we were able to utilize a data set containing three years of order-level transaction sales history for a boutique winery. The winery sales dataset includes 65,534 sales records from 22,869 unique customers from 2008 to 2010, with 21 data points per transaction. Data points include customer details such as customer ID, aggregations of sales by year, home state, ZIP code, and subscription status to email and newsletter. Order-related data points include sales by channel, order ID, date, and total sales amount.

Before starting the analysis, we reviewed the dataset to identify any cleaning or transformation needs. For instance, the state column had missing or incorrect entries that needed correction. Inconsistent totals in the Order columns, where NaN values appeared despite existing Sales data for the year, led us to focus mainly on sales volume. Additionally, the flat table structure caused redundancy of annual totals in the Sales columns for each Customer ID, requiring careful handling during summation.

Sales occur through four primary channels: Email, Newsletter, Tasting Room, and Winemaker Calls. Customers are grouped into four segments: High Roller, Casual Visitor, Wine Enthusiast, and Luxury Estate, allowing for analysis of customer behavior and performance across distinct groups.

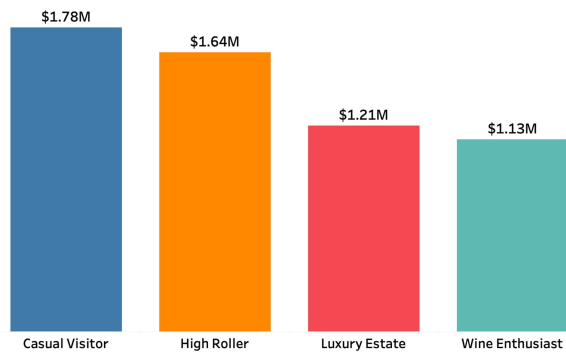


Figure 1.1 - Sum of sales by Segment

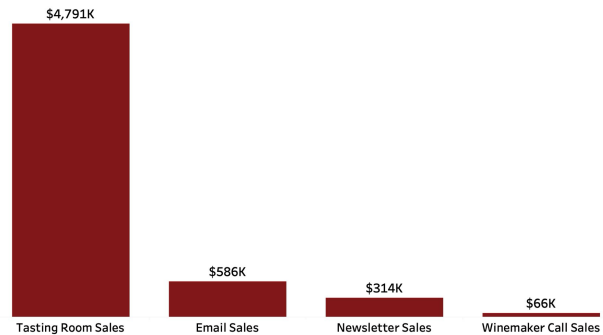


Figure 1.2 - Sum of sales by Channel

Recency, Frequency, Monetary Value (RFM) Model

Our data set was well-suited for using the RFM model because it already included all the necessary components: date, transaction count, and sales volume. Although our data set already features a Customer Segment, the RFM model provides dynamic customer labeling based on activity. This dynamic labeling enables more strategic matching of marketing channel costs to customer value.

To build the RFM model, we first grouped the data by Customer ID. We then calculated the three RFM metrics: Recency (days since the customer's most recent purchase), Frequency (number of unique orders), and Monetary (total sales amount for each customer). Below is a snippet of our Python code to illustrate this step.

```
rfm = (df.groupby("Customer_ID")
      .agg({"Date": lambda x: (snapshot_date - x.max()).days,      # Recency
           "Order_ID": "nunique",                                # Frequency
           "Sale_Amount": "sum" }) .reset_index())                # Monetary
```

After building the raw RFM table, we assigned each metric a score from 1 to 4. Recency used standard quartiles, while Frequency and Monetary were scored with a custom function that handles skewed data and duplicate bin edges. This made the scoring more reliable across uneven distributions. Once the scoring was complete, we combined the three values into a single total RFM score for each customer.

RFM Segments	Count	Avg Sales
Champions	5775	\$114.61
Loyal Customers	4927	\$89.31
Potential Loyalists	5830	\$73.10
Needs Attention	6261	\$41.40
At Risk	76	\$62.09

Finally, we used the total RFM score to place each customer into one of five segments: Champions, Loyal Customers, Potential Loyalists, Needs Attention, and At Risk.

This gave us a complete segmentation of the customer base, allowing us to compare the behaviors and values of different groups.

RFM Findings

After segmenting the customers into the five RFM groups, we were able to compare their purchasing behavior and overall value. On average, Champions spent the most per purchase at about 114 dollars. Loyal Customers spent 90 dollars, Potential Loyalists spent 73 dollars, Needs Attention spent 41 dollars, and At Risk spent 62 dollars. Champions also made purchases far more often than any other segment.

When looking at total spending across all years, the differences became even clearer. Champions spent more than 500 dollars on average. Loyal Customers spent about 250 dollars, Potential Loyalists spent about 200 dollars, Needs Attention spent about 150 dollars, and At Risk customers spent under 100 dollars. We also found important trends related to email subscriptions. For Champions, Loyal Customers, and Potential Loyalists, being an email subscriber increased their average order value. For the Needs Attention and At Risk groups, email subscription had no effect.

There were also clear patterns in purchasing channels. Winemaker Call sales only happened for Champions and Loyal Customers. Email sales accounted for 10 -15% of total spending for these higher-value groups. This indicates that boosting email-driven purchases or encouraging more personal interactions like Winemaker Calls could help turn Potential Loyalists into higher-value customer segments.

Overall, the RFM model provided us with a clearer, more helpful view of customer behavior than the original customer categories did. This will help in developing more effective and efficient marketing strategies going forward.

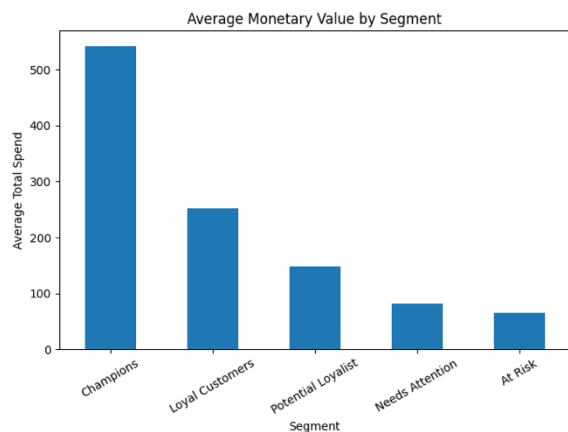


Figure 2.1 – Average lifetime spend by category

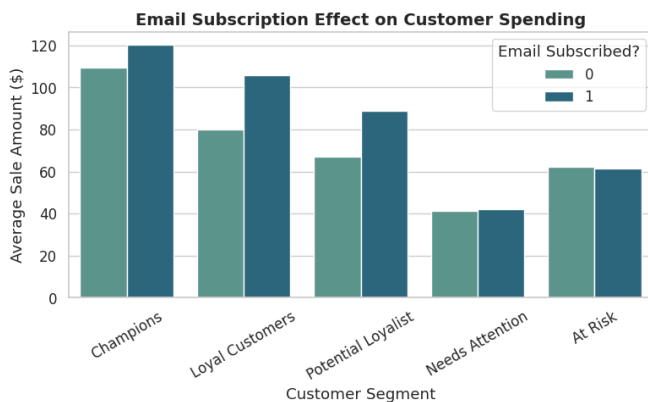


Figure 2.2 – Email effect on average purchases

Winemaker Calls: Logistic Regression Model

During our analysis of the dataset, we found that Winemaker Calls produced the highest-margin sales compared to channels like email, newsletter, and tasting room. This indicated that Winemaker Calls could be a valuable target for specific outreach efforts. The main business question was: Which customers should we focus on for Winemaker Calls? Specifically, we aimed to identify customers who hadn't had a Winemaker Call before but seemed likely to respond.

The logistic regression model is built at the customer level with one row per customer. We treated 2010 as the “future” year, using only information available by the end of 2009. For all pre-2010 customers, we created features from their 2008 and 2009 behavior, including total spending, orders, segment, state, and engagement through email, newsletter, and tasting room.

We deliberately excluded all Winemaker Call data from the feature set, so the model could not rely on past Winemaker Call activity. The target variable is binary, with 1 indicating the customer had any Winemaker Call Sales in 2010 and 0 otherwise. A logistic regression model was trained with class weights set to “balanced” because only 1.8% of customers made a Winemaker Call purchase in 2010.

Using a 70 percent training and 30 percent test split, the model outputs a probability score for each customer, which we use to rank customers from most to least likely to make a Winemaker Call purchase in 2010. In effect, the model learns from the 2010 Winemaker Call buyers, including those who used the channel for the first time, and identifies other customers whose 2008 and 2009 profiles look similar.

Findings and Interpretation

Evaluating the logistic model's ROC curve, we found that the Area Under the Curve (AUC) was 0.845 on the test set, indicating good separation between customers who made Winemaker Call purchases in 2010 and those who did not. Since Winemaker Call buyers are rare, we evaluated the model as a ranking tool rather than relying on a fixed probability threshold.

```
===== MODEL PERFORMANCE =====
ROC AUC: 0.845

Confusion matrix (rows=actual, cols=predicted):
[[5094 1269]
 [ 34   85]]

Classification report:

```

	precision	recall	f1-score	support
0	0.993	0.801	0.887	6363
1	0.063	0.714	0.115	119
accuracy			0.799	6482
macro avg	0.528	0.757	0.501	6482
weighted avg	0.976	0.799	0.872	6482

Figure 3.1 – Logistic Model Performance

The base rate in the test set is 1.8 percent, but among the top 10 percent of customers ranked by the model, the Winemaker Call buyer rate rises to 9.3 percent and this group contains 50.4 percent of all 2010 Winemaker Call buyers. The top 20 percent of customers by score still have a buyer rate of 6.5 percent and capture 70.6 percent of all buyers. In other words, the model concentrates most of the Winemaker Call demand into a relatively small slice of the customer base.

```

=== Lift vs base rate at different target sizes ===
Top 1% (n=64): buyer rate=0.156, lift=8.5x, captured buyers=10/119 (8.4% of all buyers in test)
Top 5% (n=324): buyer rate=0.117, lift=6.4x, captured buyers=38/119 (31.9% of all buyers in test)
Top 10% (n=648): buyer rate=0.093, lift=5.0x, captured buyers=60/119 (50.4% of all buyers in test)
Top 20% (n=1296): buyer rate=0.065, lift=3.5x, captured buyers=84/119 (70.6% of all buyers in test)

```

Figure 3.2 – Logistic model findings

The “top 20 customers to target” list illustrates how this ranking can be used in practice when calling capacity is very limited. These 20 customers are not simply the people who already bought through Winemaker Calls. They are the customers whose 2008 and 2009 behavior makes them most similar to the group of customers who bought via Winemaker Calls in 2010, including many first-time Winemaker Call users.

===== TOP 20 CUSTOMERS TO TARGET =====			
	proba_win_2010	target_win_2010	Customer Segment
Customer ID			
12	1.000	0	Casual Visitor
2779	0.998	0	High Roller
10030	0.998	0	High Roller
2733	0.998	0	High Roller
8296	0.998	0	High Roller
2627	0.997	0	High Roller
3613	0.997	0	High Roller
14510	0.997	0	High Roller
2707	0.997	0	High Roller
3246	0.996	0	High Roller
10111	0.996	0	High Roller
3686	0.995	0	High Roller
14166	0.995	0	High Roller
3273	0.995	0	High Roller
1153	0.994	0	High Roller
2822	0.994	0	High Roller
554	0.993	0	High Roller
10007	0.993	0	High Roller
18570	0.993	0	High Roller
2636	0.993	0	High Roller

Figure 3.3 – Customer rankings by lift

If the winery can make only a few calls at the beginning of 2010, this list will be the top priority. More broadly, the model provides a data-driven call list where the top 10 to 20 percent of customers by score include a large share of true Winemaker Call buyers. This makes Winemaker outreach more focused and efficient, while we note that the model is predictive rather than causal and does not prove that a call itself caused the purchase. The analysis revealed a clear "Green Light" target in the "High Roller" segment, who are 5.6 times more likely to purchase than the average customer, particularly when they have purchased recently.

Conversely, the model exposed a critical "Red Light" in our current strategy: customers who already subscribe to our email or newsletter are 91% less likely to buy over the phone, indicating a strong preference for digital channels over direct sales calls. Furthermore, traditional metrics like total historical spend proved to be insignificant predictors, debunking the myth that our biggest spenders are always our best targets for every channel. Implementing these findings, prioritizing High Rollers and removing digital subscribers from call lists, will significantly increase the efficiency and ROI of the program.

Logit Regression Results						
Dep. Variable:	Target	No. Observations:	8796			
Model:	Logit	Df Residuals:	8787			
Method:	MLE	Df Model:	8			
Date:	Sat, 06 Dec 2025	Pseudo R-squ.:	0.3779			
Time:	02:40:36	Log-Likelihood:	-1149.9			
converged:	True	LL-Null:	-1848.3			
Covariance Type:	nonrobust	LLR p-value:	2.758e-296			
	coef	std err	z	P> z	[0.025	0.975]
const	0.4009	0.185	2.163	0.031	0.038	0.764
Recency	-0.0122	0.001	-16.762	0.000	-0.014	-0.011
Frequency	-0.0137	0.026	-0.525	0.600	-0.065	0.037
Monetary	-3.592e-05	3.08e-05	-1.167	0.243	-9.62e-05	2.44e-05
Email Subscr	-2.4143	0.161	-14.960	0.000	-2.731	-2.098
Newsletter Subscr	-0.5914	0.166	-3.569	0.000	-0.916	-0.267
Customer Segment_High Roller	1.7180	0.162	10.612	0.000	1.401	2.035
Customer Segment_Luxury Estate	-55.6403	1.23e+11	-4.51e-10	1.000	-2.42e+11	2.42e+11
Customer Segment_Wine Enthusiast	-0.0053	0.188	-0.028	0.978	-0.375	0.364

Figure 3.4 – Logit regression results

Customer Churn Analysis: Logistic Regression

To accurately determine the true drivers of customer retention, we designed a logistic regression model that functions as a controlled experiment rather than a simple comparison. We restricted our analysis to a specific cohort of customers acquired prior to 2010, defining "churn" strictly as zero spending activity throughout the entire 2010 fiscal year.

Critically, to ensure scientific validity, we included control variables for "Customer Segment" (wealth) and "Past Spend" (historical value). This methodology allowed us to neutralize the inherent bias that wealthier customers naturally subscribe more often, enabling us to isolate the pure causal impact of email subscription on retention.

The results uncovered a counterintuitive "Retention Paradox." Contrary to the standard assumption that digital engagement fosters loyalty, our model indicates that email subscribers are statistically approximately 12% *more* likely to churn than non-subscribers when holding customer type constant (Coefficient +0.11, P-Value = 0.010). The analysis reveals that the strong retention rates we typically associate with subscribers are actually driven by the customers' inherent characteristics—specifically, being in the "High Roller" or "Wine Enthusiast" segments—rather than by the communication they receive. In short, these customers stay because of who they are, not because of the emails they read.

Furthermore, the model debunked a significant myth regarding customer value: "Past Spend" proved to be statistically insignificant (P = 0.29), meaning a customer's financial history is not a reliable predictor of their future loyalty. This finding, combined

with the negative impact of subscriptions, suggests a critical flaw in our current engagement strategy. We are not losing customers despite our emails; the data implies we may be losing customers *because* of them. Consequently, the immediate strategic recommendation is to move beyond simple email capture and urgently overhaul the content strategy to ensure it delivers genuine value rather than driving attrition.

Churn Analysis Results:

Logit Regression Results									
Dep. Variable:	Churned	No. Observations:	21046						
Model:	Logit	Df Residuals:	21040						
Method:	MLE	Df Model:	5						
Date:	Sat, 06 Dec 2025	Pseudo R-squ.:	0.08278						
Time:	19:17:52	Log-Likelihood:	-11371.						
converged:	True	LL-Null:	-12397.						
Covariance Type:	nonrobust	LLR p-value:	0.000						
	coef	std err	z	P> z	[0.025	0.975]			
const	1.5874	0.027	58.187	0.000	1.534	1.641			
Email Subscr	0.1113	0.043	2.589	0.010	0.027	0.196			
Past_Spend	-2.431e-05	2.31e-05	-1.053	0.292	-6.96e-05	2.09e-05			
Customer Segment_High Roller	-1.9893	0.057	-34.950	0.000	-2.101	-1.878			
Customer Segment_Luxury Estate	-0.5158	0.041	-12.498	0.000	-0.597	-0.435			
Customer Segment_Wine Enthusiast	-1.4921	0.052	-28.690	0.000	-1.594	-1.390			

Clean Summary Table				
	Feature	Coefficient	P-Value	Odds Ratio \
0	const	1.587401	0.000000e+00	4.891019
1	Email Subscr	0.111290	9.628526e-03	1.117719
2	Past_Spend	-0.000024	2.923396e-01	0.999976
3	Customer Segment_High Roller	-1.989308	1.298295e-267	0.136790
4	Customer Segment_Luxury Estate	-0.515781	7.638871e-36	0.597034
5	Customer Segment_Wine Enthusiast	-1.492089	5.077746e-181	0.224902

Figure 4.1 – Churn analysis logit regression results

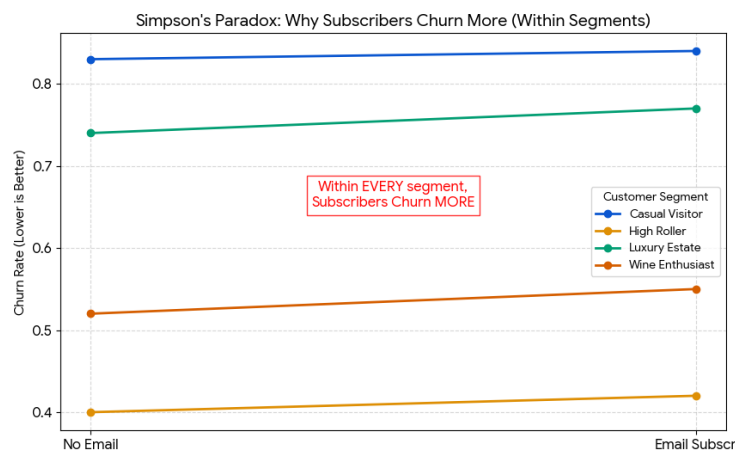


Figure 4.2 – Churn rate by customer segment

Customer Lifetime Value: Linear Regression

To address the potential "correlation vs. causation" skepticism—specifically, whether multi-channel customers spend more simply because they have been customers longer—we conducted a rigorous linear regression analysis to isolate the true drivers of Customer Lifetime Value (LTV). This model was designed to quantify the specific "Omnichannel Effect" by mathematically controlling for "Tenure Years" (time bias) and "Customer Segment" (wealth bias). This approach allowed us to separate the value generated by the strategy itself from the inherent value of the customer's loyalty or spending power.

The results confirm a highly significant "Omnichannel Premium" of \$44.62 per customer (P-Value = 0.006), proving that cross-channel engagement creates incremental value independent of how long a customer stays. To put this in strategic perspective, the model shows that one year of customer retention adds approximately \$15 to LTV; therefore, converting a customer to a second channel is the revenue equivalent of retaining them for three full years. While we cannot change a customer's inherent wealth (with segments like High Rollers naturally adding ~\$479), we *can* actively influence their channel usage. Consequently, incentivizing single-channel customers to adopt a second channel represents our most efficient lever for immediate revenue growth

OLS Regression Results						
Dep. Variable:	Sale Amount	R-squared:	0.025			
Model:	OLS	Adj. R-squared:	0.025			
Method:	Least Squares	F-statistic:	116.0			
Date:	Sat, 06 Dec 2025	Prob (F-statistic):	1.54e-121			
Time:	03:39:51	Log-Likelihood:	-1.8934e+05			
No. Observations:	22869	AIC:	3.787e+05			
Df Residuals:	22863	BIC:	3.787e+05			
Df Model:	5					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
const	140.5190	10.933	12.853	0.000	119.090	161.948
Is_Multi_Channel	44.6243	16.207	2.753	0.006	12.857	76.391
Tenure_Years	14.6890	5.294	2.774	0.006	4.312	25.066
Customer_Segment_High_Roller	479.2455	21.674	22.111	0.000	436.763	521.728
Customer_Segment_Luxury_Estate	42.7743	16.940	2.525	0.012	9.571	75.977
Customer_Segment_Wine_Enthusiast	100.9192	19.998	5.047	0.000	61.723	140.116
Omnibus:	48920.291	Durbin-Watson:	0.138			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	283766870.543			
Skew:	18.932	Prob(JB):	0.00			
Kurtosis:	547.396	Cond. No.	8.52			

Figure 5.1 – Linear regression results

Recommendations

Our analysis of winery sales, customer segments, and engagement channels provides a clear roadmap for improving marketing decisions and unlocking greater customer value. The findings highlight the importance of prioritizing high-value segments and channels, reducing churn, and building multi-channel engagement strategies that maximize customer lifetime value.

Tasting Rooms as the Anchor Channel: The Tasting Room consistently delivers the highest sales values and should remain the centerpiece of customer engagement. It provides the strongest opportunities for upselling and building emotional connections with customers. We recommend continuing to host regular events such as seasonal releases, pairing dinners, and other community engagement opportunities while using geo-targeted outreach to increase attendance from nearby zip codes. Staff should be trained to capture customers' email addresses during visits to the Tasting Room and to encourage repeat engagement.

Winemaker Calls as a Growth Segment Strategy: Rather than reserving Winemaker Calls only for top-tier customers, we recommend investing them in loyal customers who show potential to move into higher-value segments. These customers already demonstrate consistent engagement but have not yet reached the highest spending levels. Personalized winemaker calls can deepen relationships, encourage larger purchases, and accelerate their progression into more valuable segments. This positions Winemaker Calls as a strategic growth lever, not just a premium touchpoint.

Email Strategy Streamlined and Supportive: Currently, our data shows that email engagement is not strongly correlated with higher customer value, and in fact, subscribers are more likely to churn. Email campaigns should be streamlined and selective, focusing on delivering clear value like event invitations, tasting room promotions, or exclusive updates. Additionally, personalization by segment and journey stage is critical: early-stage customers may receive welcome content and simple offers, while long-term customers receive targeted updates tied to their past preferences. Email should serve as a supportive channel that reinforces other touchpoints.

Driving Multi-Channel Engagement: Customers who engage across three or more channels (tasting room, email, newsletters, winemaker calls) consistently deliver the highest lifetime value. Integrated campaigns should be designed to connect these touchpoints. For example, newsletters promoting tasting room events, tasting room staff capturing email sign-ups, and winemaker calls inviting VIPs to private tastings. Measuring and optimizing cross-channel engagement will ensure that customers move up the value ladder, increasing both retention and profitability.

Future Analysis

A high-level review of total sales and distinct customer counts by state reveals several insights that warrant deeper modeling to inform a data-driven strategy. In some states, customer counts are relatively large, but overall sales remain moderate, highlighting opportunities to increase average sales amount and strengthen loyalty. Conversely, other states show relatively low customer counts yet deliver high sales volumes, suggesting strong potential to identify and cultivate additional higher-value lookalike customers. These patterns point to distinct pathways for growth: expanding wallet share in broad but underperforming markets, while replicating high-value behaviors in smaller but more profitable regions.

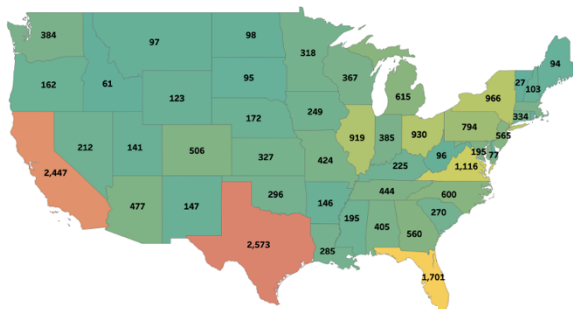


Figure 6.1 – Distinct customers by state

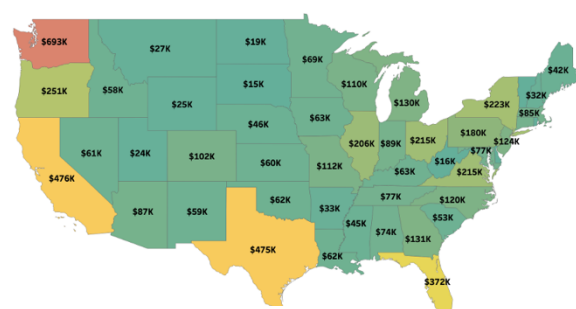


Figure 6.2 – Total sales per state

Looking ahead, a future analysis project could focus on geo-targeting by zip code. Mapping customer density and performance by location will allow the winery to reinforce loyalty in high-performing areas and convert high-potential regions into growth markets. This approach ensures marketing resources are allocated efficiently, driving both immediate sales and long-term growth.

- **Objective:** Identify high-performing areas for retention and upsell, and high-potential areas for acquisition.
- **Approach:**
 - Map customer density and sales performance by zip code.
 - Segment regions into *high-performing* (loyal, repeat buyers) and *high-potential* (large populations, moderate engagement).
 - Design targeted campaigns for each region — exclusive tasting room invitations for nearby customers, digital ads for emerging markets, and local partnerships to expand reach.
- **Expected Outcomes:**
 - More efficient marketing spend by focusing resources where ROI is highest.
 - Increased tasting room traffic and wine club sign-ups in strong markets.
 - Expanded penetration into new regions with untapped potential.

Conclusion

In summary, the path forward is clear: continue leveraging the tasting room as the anchor channel, invest Winemaker Calls in loyal customers with growth potential, revamp the email strategy to reduce churn and drive measurable outcomes, and build multi-channel engagement strategies. With geo-targeting as the next step, the winery can sharpen outreach and ensure growth in both established and emerging markets, truly unlocking a greater share of customer value.